

Vidyut Sriram

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<https://arxiv.org/abs/2601.00509>

OBJECTIVE

Computer science student with experience in backend systems and AI-assisted developer tooling, focused on building reliable, secure, and scalable software.

EDUCATION

Pennsylvania State University, University Park, PA

Aug 2023 - May 2027

Bachelor of Science, Computer Science, Minor in Math, Artificial Intelligence and Engineering Leadership Development

- **GPA:** 3.85/4
- **Achievements:** Dean's List 6/6 semesters

WORK EXPERIENCE

Comcast India Engineering Centre

Software Development Engineer Intern

Chennai, TN · May 2025 - Aug 2025

- Implemented an agentic AI system using RAG, modular pipelines, and multi-step reasoning; integrated external APIs and data sources to automate code evaluation and generation workflows, reducing manual coding effort by 40% and improving pipeline reliability.
- Built a personalized chatbot interface for document-based Q&A and task automation, improving workflow efficiency by 18% through context-aware and scalable automation.

Pennsylvania State University

Learning Assistant

PA · Aug 2025 - Present

- Facilitated weekly review sessions implementing active learning strategies in collaboration with faculty, driving measurable improvements in student grades, participation, and comprehension of limits, derivatives, and integrals.

Math Peer Tutor

PA · Jan 2026 - Present

- Supported students across 0–100 level math courses through one-on-one and small-group tutoring, clarifying foundational concepts, guiding practice problems, and building confidence and effective study habits.

PROJECTS

LLM-Assisted Secure Code Generation with Automated Feedback

Sep 2025 - Mar 2026

Penn State Undergraduate Exhibition for Research, Inquiry, and Creative Activity 2026
Place, Engineering

2nd Pennsylvania State University

- Built a RAG-augmented secure code generation pipeline combining GCC compilation checks, GitHub CodeQL static analysis, and KLEE symbolic execution to automatically detect and repair security vulnerabilities in LLM-generated C programs.
- Designed an iterative repair loop where structured diagnostic feedback is fed back to the model across multiple attempts, reducing compilation failures by ~50% and security defects by ~55% for DeepSeek Coder 1.3B on a dataset of 5,000 C programming tasks.

PrismAI - AI Meeting Intelligence Platform

Apr 2026 - Present

- Built a multi-user meeting intelligence platform where Recall.ai bots join Zoom/Meet/Teams calls and 8 specialized AI agents on a two-tier LangGraph pipeline generate summaries, action items, decisions, sentiment analysis, speaker coaching, and follow-up emails in ~2s via SSE streaming.
- Developed a workspace-shared RAG system with PDF/Notion/Drive ingestion, automatic transcript indexing, hybrid retrieval (pgvector + BM25 + RRF), LLM reranking, synced meeting playback, and Slack/Notion export. Stack: React, FastAPI, LangGraph, Groq Llama 3.3-70B, OpenAI, Supabase/pgvector, Recall.ai, Tavily.

Terrametric - Crop Drought Stress Intelligence Platform

Apr 2026

1st Place Real World Track (Ambrook) & Data Science Track, 3rd Place Open Source - BitCamp 2026, UMD (36-hour hackathon)

- Built a field-level crop stress intelligence platform that scores drought risk, generates actionable irrigation prescriptions with cost/savings projections, and lets farmers simulate decisions before acting - bridging the gap between raw satellite/climate data and on-farm decisions.
- Developed a 14-day per-field stress forecast, a seasonal harvest outlook, and an agentic AI layer (Featherless AI / Qwen 2.5-7B) with RAG over Iowa/USDA documents; built a React globe/map frontend (Three.js, Leaflet, Recharts) with FastAPI backend, XGBoost ML pipeline, Twilio SMS, and WebSocket streaming deployed on Render/Vercel.

SKILLS

- **Languages:** Python, Java, C, C++, SQL, JavaScript
- **Frontend:** React, Three.js, React Three Fiber, Tailwind CSS, Vite
- **Backend:** REST APIs, Debugging, Git, Pipeline Orchestration, Docker, FastAPI, Websockets, SSE, SQLite, PostgreSQL
- **AI/ML:** Langflow, Prompt Engineering, Retrieval-Augmented Generation, XGBoost, Scikit-learn, Tool-Calling, Groq, OpenAI API
- **Tools:** CodeQL, KLEE, n8n, Supabase, Render, Railway, Vercel, Recall.ai, ElevenLabs, Tavily